P395 Directed/Self-directed Problems

For the last couple weeks of the course, you can choose to do **two** directed problems from the three choices that I provide guides for, or study a single numerical problem of your own choosing. If you are considering studying your own problem, it is vital that you consult with me about the suitability of the material and its scope.

Organization and Writeup:

By the due date, you will submit a Jupyter notebook containing your work for the given problem. Your notebook should be organized with the following sections:

1. Introduction
2. Model/Methods
3. Results/Analysis
4. Discussion/Conclusions

Here is a brief summary of what each section should contain:

<u>Introduction</u>: A brief summary in your own words of the problem, stating the question(s) that you will address with your computational analysis. It should cite any necessary references.

<u>Theory/Methods</u>: Describe the model(s) you will be simulating/analyzing. You should present the equations as well as a description of the computational methodology you use to simulate them. It doesn't have to be exhaustive, but I should be able to know the system you are studying and the computational approach. If several methods are being used, you might find it helpful to break it up into meaningfully labeled subsections.

<u>Results/Analysis</u>: Your code along with the associated generated graphs/plots. It should be readable and executable and contain enough text to connect one part to the other so that I can logically follow what you are doing. You might find it useful to break this section up into meaningfully labeled subsections.

<u>Discussions/Conclusions</u>: Summarize your findings. Were you able to answer your questions? This is also the place to discuss potential future directions for how you could change the model or study some other aspect of it.

Marking scheme

I will use the following rubric:

Grading scheme (points): 1 = poor/unacceptable/many errors 2 = average/some errors

3 = good/very few errors 4 = excellent/error free

Category	Points	Weight
Clarity/Readability: • organized? • clear introduction/methods/equations? • presentation quality?	4	30%
Results/Analysis • plot quality? • results scientifically sound/error free? • technically correct?	4	30%
Coding: • code readable? • code runs?	4	30%
Conclusions/discussion: • conclusions reasonable? • future directions feasible/sensible?	4	10%